

CABO CARVOEIRO, Portugal

WMO: 085310

Lat: 39.3614N

Lon: 9.4069W

Elev: 32

StdP: 100.94

Time Zone: 0.00 (EUW)

Period: 90-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.6	7.8	-4.2	2.7	11.7	-0.6	3.6	9.6	14.8	11.9	13.2	10.4	5.5	90	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.0	23.3	18.6	22.1	18.8	21.2	18.6	20.3	21.8	19.7	21.1	19.3	20.6	3.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.7	14.5	21.1	19.2	14.0	20.6	18.7	13.6	20.2	58.4	21.9	56.5	21.2	55.0	20.8	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	11.0	9.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	12.4	12.3	13.6	14.6	16.3	17.8	18.8	19.0	18.7	17.6	15.0	12.9
	DBStd	2.99	1.87	1.86	1.77	1.90	1.73	1.28	1.11	1.14	1.32	1.80	2.05	1.96
	HDD10.0	12	4	4	0	0	0	0	0	0	0	0	0	3
	HDD18.3	1029	183	169	147	114	67	26	6	3	11	35	101	167
	CDD10.0	2119	79	69	112	138	196	233	274	279	261	235	149	94
	CDD18.3	96	0	0	1	2	4	8	22	25	22	12	1	0
	CDH23.3	47	0	0	1	2	8	1	7	12	6	11	0	0
	CDH26.7	4	0	0	0	0	1	0	0	2	1	1	0	0
Wind	WSAvg	4.9	5.2	5.6	5.8	5.2	4.7	4.6	4.6	4.6	4.3	4.2	5.2	5.3
Precipitation	PrecAvg	599	84	75	45	52	42	17	5	6	24	71	91	86
	PrecMax	897	264	244	137	138	118	66	31	51	96	194	288	261
	PrecMin	366	1	6	0	8	0	0	0	0	1	2	1	6
	PrecStd	143	54	52	35	32	33	19	7	10	25	56	61	65
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.6	17.8	21.6	23.0	23.9	23.0	24.1	24.4	24.0	24.8	20.3	17.7
		MCWB	14.3	13.3	14.2	15.8	17.7	19.3	20.4	20.0	19.7	16.9	16.5	15.2
	2%	DB	15.8	16.1	18.4	20.1	21.0	21.2	22.1	22.4	22.2	21.9	19.1	16.7
		MCWB	13.6	13.3	13.9	16.2	17.2	18.4	19.5	19.8	19.4	17.3	16.7	15.0
	5%	DB	15.3	15.3	16.8	18.2	19.5	20.2	21.1	21.4	21.3	20.7	18.3	16.2
		MCWB	13.2	13.2	13.5	15.6	16.6	17.8	19.0	19.2	19.1	17.7	16.1	14.5
	10%	DB	14.9	14.7	15.8	17.0	18.5	19.6	20.4	20.7	20.6	19.9	17.7	15.6
		MCWB	12.8	12.9	13.2	15.0	16.2	17.3	18.5	18.8	18.8	17.7	15.5	13.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	15.1	16.2	18.0	19.1	19.9	21.0	21.4	21.0	19.9	17.9	16.3
		MCDB	16.1	16.0	18.4	20.7	21.6	22.2	22.8	23.3	23.0	20.9	18.9	17.0
	2%	WB	14.8	14.4	15.1	16.5	17.9	19.0	20.1	20.2	20.1	19.2	17.1	15.6
		MCDB	15.5	15.5	16.6	18.7	19.7	20.7	21.4	21.6	21.4	20.6	18.2	16.4
	5%	WB	14.2	13.8	14.4	15.8	17.1	18.2	19.4	19.6	19.5	18.6	16.4	15.1
		MCDB	15.1	15.0	15.9	17.9	18.9	19.7	20.8	20.9	20.8	20.1	17.6	15.9
	10%	WB	13.6	13.2	13.7	15.1	16.5	17.7	18.7	19.1	19.1	18.0	15.8	14.5
		MCDB	14.6	14.5	15.3	16.9	18.3	19.2	20.2	20.3	20.4	19.4	17.1	15.5
Mean Daily Temperature Range	5% DB	MDBR	3.1	3.4	3.4	3.1	2.9	2.6	2.7	3.0	2.9	3.6	3.2	3.5
		MCDBR	3.2	4.4	6.1	5.6	5.2	3.9	3.9	4.4	4.2	5.4	3.5	3.6
		MCWBR	3.0	3.4	4.2	3.3	2.9	2.3	2.4	2.5	2.7	3.6	3.5	3.4
	5% WB	MCDBR	2.4	3.4	4.3	4.8	4.4	3.6	3.6	3.7	4.0	4.5	2.9	2.8
		MCWBR	2.7	3.1	3.8	3.1	2.8	2.4	2.6	2.5	2.8	3.2	3.0	3.3
Clear-Sky Solar Irradiance	taub	0.334	0.345	0.376	0.382	0.389	0.394	0.381	0.392	0.391	0.369	0.343	0.333	
	taud	2.506	2.472	2.393	2.382	2.394	2.399	2.430	2.408	2.423	2.491	2.517	2.505	
	Ebn at Noon	835	876	879	894	891	883	891	873	853	837	817	806	
	Edh at Noon	78	91	109	117	118	118	113	113	105	89	76	73	
All-Sky Solar Radiation	RadAvg	2.12	3.13	4.28	5.58	6.63	7.03	7.09	6.53	5.10	3.53	2.38	1.92	
	RadStd	0.27	0.36	0.45	0.46	0.47	0.34	0.33	0.27	0.32	0.26	0.29	0.21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page